

National University of Computer and Emerging Sciences
Lahore Campus

Computer Networks (CS3001)

Date: February 23rd 2026

Course Instructor(s)

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Sessional-I Exam

Total Time: 1 Hours

Total Marks: 45

Total Questions: 5

Roll No

Section

Student Signature

- Instruction/Notes:**
- Attempt all questions on the provided separate answer sheet.
 - Clearly write corresponding question number and part number at the top center of the answer sheet with a thick pen / marker before starting a new question / answer.
 - In case you have used rough sheets, they should **NOT** be attached to the answer sheet.

CLO 1 (Q1 to Q2): Describe utilization of network protocol concepts vis-a-vis OSI and TCP/IP stack.

[10 + 10 = 20 Marks]

Q1 (a): A source host sends a message $M = 600,000$ bits to a destination host through two intermediate routers (a total of 3 hops). Each link has a transmission rate of $R = 10$ Mbps (where $1 \text{ M} = 10^6$). The propagation delay for each link is 10 ms (where $1 \text{ ms} = 10^{-3}$). The message is segmented into 300 packets by the source host, each of length $L = 2,000$ bits. Assuming store-and-forward switching and zero processing or queuing delays, what is the total end-to-end delay to deliver the entire message to the destination? **[5 Marks]**

Answer: First, we find the transmission and propagation delay for one 2,000-bit packet over first link.

- Transmission Delay = $L / R = 2,000 / 10,000,000 = 0.0002 \text{ s} = 0.2 \text{ ms}$
- Propagation Delay = 10 ms

So total time for 1st packet = transmission delay 1st link + propagation delay 1st link + transmission delay 2nd link + propagation delay 2nd link + transmission delay 3rd link + propagation delay 3rd link = $3 * \text{transmission delay} + 3 * \text{propagation delay} = (3 * 0.2 \text{ ms}) + (3 * 10 \text{ ms}) = 30.6 \text{ ms}$

The destination receives one new packet after every transmission delay interval, after the first packet arrives. Thus 299 packets remain. So additional time required = $299 * \text{transmission delay} = 299 * 0.2 \text{ ms} = 59.8 \text{ ms}$

Total time = $30.6 \text{ ms} + 59.8 \text{ ms} = 90.4 \text{ ms}$

(b): For this part, assume that the message M in part (a) is not segmented into smaller packets but is instead sent as one big packet of length $L = 600,000$ bits by the source host. Would the total delay for this packet to reach the destination: remain the same, increase or decrease compared to your answer in part (a)? Justify your answer. (Assume everything else is the same as in part (a).) **[5 Marks]**

Answer: Total delay increases.

Justification: Transmission Delay = $L / R = 600,000 / 10,000,000 = 60 \text{ ms}$

Total delay = $3 * \text{transmission delay} + 3 * \text{propagation delay} = (3 * 60) + (3 * 10) = 210 \text{ ms}$

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Q2: A client downloads 2000 bytes of App layer data from a server using the TCP/IP Layered Model. Assume:

- Application layer adds no header
- Transport Layer and Network Layer add a 20-byte header each
- Link Layer adds a total of 22 bytes (header + trailer)
- The entire data fits into one transport layer segment
- The client and server are connected via one switch and one router only

Answer the following questions:

[1 * 10 Marks]

- What is the total size of the transport layer segment payload? **2000 bytes**
- What is the total size of the transport layer segment? **2020 bytes**
- What is the total size of the network layer datagram? **2040 bytes**
- What is the total size of the link layer frame? **2062 bytes**
- What is the total overhead percentage? **$(62 / 2062) * 100 = 3\%$**
- Which layer is responsible for enabling process to process communication? **Transport Layer**
- Which layer is responsible for host-to-host delivery? **Network Layer**
- Which layer is responsible for local delivery? **Link Layer**
- Which layer(s) are not visible to the switch? **Network, Transport & Application Layers**
- Which layer(s) are not visible to the router? **Transport & Application Layers**

CLO 2 (Q3 to Q5): Demonstrate the basics of network concepts using state-of-the-art network tools/techniques.

[14 + 5 + 6 = 25 Marks]

Q3 (a): A client retrieves a Web page consisting of one base HTML file and 6 referenced objects, whereas the propagation delay = 30 ms, thus 1 RTT = 60 ms. DNS lookup requires two DNS servers and iterative queries are used, with each DNS query taking 1 RTT (i.e. total time for one DNS query including request & response.) The base object size is 120K Bytes, while each referenced object is 10 K Bytes & the link transmission rate R = 10 Mbps. Compute the total delay to retrieve the entire webpage for the following cases:

[2 * 3 + 4 = 10 Marks]

- Non-persistent HTTP with no parallel connections
- Persistent HTTP without pipelining (no parallel connections)
- Persistent HTTP with pipelining (no parallel connections)
- Non-persistent HTTP with maximum three parallel connections possible (no pipelining)

Solution:

RTT = 60 ms

T.D base file = $120 \text{ K} * 8 / 10 \text{ Mbps} = 96\text{ms}$

T.D per ref file = $10 \text{ K} * 8 / 10 \text{ Mbps} = 8 \text{ ms}$

(i) DNS delay = 2 RTT = $2 * 60 = 120 \text{ ms}$

Total Delay = DNS delay + 7 * (1 RTT for TCP + 1 RTT to request file) + T.D of base + TD of all 6 ref. files =
DNS delay + 14 RTT + T.D of base + TD of all 6 ref. files = $120 \text{ ms} + 7 * (2 * 60 \text{ ms}) + 96 \text{ ms} + (6 * 8 \text{ ms}) =$
1104 ms

(ii) DNS delay = 2 RTT = $2 * 60 = 120 \text{ ms}$

Total Delay = DNS delay + 1 RTT for TCP + 7 RTT to request all 7 files + T.D of base + TD of all 6 ref. files
= $120 \text{ ms} + 60 \text{ ms} + (7 * 60\text{ms}) + 96 \text{ ms} + (6 * 8 \text{ ms}) =$ **744 ms**

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(iii) DNS delay = 2 RTT = $2 * 60 = 120$ ms

Total Delay = DNS delay + 1 RTT for TCP + 1 RTT to request base file + 1 RTT to request all 6 ref. files + T.D of base + TD of all 6 ref. files = DNS delay + 3 RTT + T.D of base + TD of all 6 ref. files = 120 ms + $(3 * 60$ ms) + 96 ms + $(6 * 8$ ms) = **444 ms**

(iv) DNS delay = 2 RTT = $2 * 60 = 120$ ms

Total Delay = DNS delay + 1 RTT for TCP + 1 RTT to request base file + T.D of base file + 1 RTT to open 3 parallel connections + 1 RTT to request 3 parallel ref. files + 1 T.D for 3 ref. files in parallel + 1 RTT to open 3 parallel connections + 1 RTT to request 3 parallel ref. files + 1 T.D for 3 ref. files in parallel = DNS delay + 6RTT + T.D of base + 2 T.D of ref. file = 120 ms + $(6 * 60$ ms) + 96 ms + $(2 * 8$ ms) = **592 ms**

Q3 (b): A client is connected to a local proxy cache, which in turn is connected to an origin server with the following characteristics:

- $RTT_{cl} = 10$ ms (i.e. Round-Trip Time between the client & the local cache)
- $RTT_{ls} = 80$ ms (i.e. Round-Trip Time between the local cache & the origin server)
- $T_{trans} = 30$ ms (i.e. Time to transmit the object)
- Ignore any other delays

Assume that persistent TCP connections are already established to both the cache and the server. The client requests an object that is already stored in the local proxy cache. Calculate the total response time for the following two scenarios:

[2 + 2 = 4 Marks]

- i) Scenario A: The Object in the cache is fresh.
- ii) Scenario B: The Object in the cache is stale.

Solution:

- i) $RTT_{cl} + RTT_{ls} + T_{trans} = 10 + 80 + 30 = 120$ ms
- ii) $RTT_{cl} + RTT_{ls} + 2 * T_{trans} = 10 + 80 + (2 * 30) = 150$ ms

Q4: An email of 20 KB is sent from a client to a mail server via SMTP. The propagation delay = 20 ms, link bandwidth = 2 Mbps, and SMTP requires 4 round-trip command sequences (HELO, MAIL FROM, RCPT TO, DATA/QUIT). Compute approximate total time to send the email (ignoring server processing). **[5 Marks]**

Solution

Step 1 – Convert email size to bits:

$$20 \text{ KB} = 20 \times 1024 \times 8 = 163840 \text{ bits}$$

Step 2 – Transmission time:

$$\text{time}_{\text{transmit}} = 163840 \text{ bits} / 2 \times 10^6 \text{ bps} = 0.08192 \text{ s} \approx 82 \text{ ms}$$

Step 3 – Total propagation for 4 round-trips:

$$p_{\text{time}} = 4 * 20 \text{ ms} * 2 \text{ (round-trip)} = 160 \text{ ms}$$

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Step 4 – Total time:

$$T_{\text{total}} = \text{time_transmit} + p_time = 82 + 160 \approx 242 \text{ ms}$$

Q5: A user requests `www.university.edu.pk`. The local DNS server caches nothing. The delays are:

- Local Host to Local DNS: 2 ms (One way).
- Local DNS to Root Server: 40 ms (RTT).
- Local DNS to .pk TLD Server: 35 ms (RTT).
- Local DNS to .edu.pk Authoritative Server: 20 ms (RTT).
- Local DNS to university.edu.pk Server: 15 ms (RTT).

The DNS resolution is Iterative. The RTT between the client and the web server is 50 ms. Assume a new TCP connection is established and requires one RTT for the TCP handshake and one RTT for the HTTP request/response. Calculate the total time from clicking the link to receiving the initial HTTP response, assuming the Local DNS server must contact Root, then TLD, then Authoritative. The local DNS server must also query the final authoritative server for `university.edu.pk` to obtain the IP address. **[6 Marks]**

Solution:

1. DNS Resolution Time:

$$T_{\text{DNS}} = 2 \text{ (Host} \rightarrow \text{DNS)} + 40 \text{ (Root)} + 35 \text{ (TLD)} + 20 \text{ (Auth)} + 15 \text{ (Server)} + 2 \text{ (DNS} \rightarrow \text{Host)}$$

$$T_{\text{DNS}} = 114 \text{ ms}$$

2. HTTP Connection & Response:

$$T_{\text{HT T P}} = 1 \times \text{RTT}_{\text{HTTP (TCP Handshake)}} + 1 \times \text{RTT}_{\text{HTTP (HTTP Request)}} = 2 \times 50 = 100 \text{ ms}$$

3. Total Time:

$$\text{Total} = 114 \text{ ms} + 100 \text{ ms} = 214 \text{ ms}$$